Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

Python Programming for Finance

TGS Ref No: TGS-2026065049

Conducted by

Tertiary Infotech Academy Pte Ltd

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	23 August 2026	Initial CASL/WSQ release aligned to the seven course topics, 17 realistic finance labs and current assessment design.	Dr. Alfred Ang

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Course Overview

This three-day course develops practical Python skills for finance decisions. Learners progress from requirements and programming fundamentals through reusable functions, Pandas data preparation, aggregation, visualisation and explainable statistical analysis.

Course Code	TGS-2026065049
Duration	3 days · 24 training hours
Daily Time	9:00 AM – 6:00 PM
Assessment	2 hours on Day 3: 60-minute WA + 60-minute PP
Trainer	Dr. Alfred Ang
Mode	Instructor-led concepts, demonstrations and 17 hands-on finance labs

Learning Outcomes

- LO1: Translate finance business requirements into clear Python programming objectives and configure a reproducible development environment.
- LO2: Use Python data types and operators to represent, calculate and present financial information accurately.
- LO3: Organise finance logic with conditional statements, loops and comprehensions to solve repeatable analytical problems.
- LO4: Create reusable functions and lambda-based transformations for finance business use cases.
- LO5: Import, inspect, filter, slice and clean finance data with Pandas Series and DataFrames.
- LO6: Join, aggregate, test and visualise finance data using groupby, pivot tables and appropriate charts.
- LO7: Improve analytical code with apply and pipe, use descriptive statistics, and analyse changes in financial data over time.

Daily Schedule

Day 1 — Python foundations and finance logic

Time	Duration	Topic / Activity	Method	Slides
9:00 AM–9:30 AM	30 min	Welcome, learner introductions, administration and AM digital attendance	Facilitation and administration	—
9:30 AM–10:45 AM	75 min	Topic 1 — Introduction to Python Programming	Concepts, demo and discussion	Slides 24–42
10:45 AM–11:00 AM	15 min	Morning break	Break	—
11:00 AM–1:00 PM	120 min	Labs 1–2 — environment, requirements and SGX price tool	Guided hands-on and evidence review	Slides 36–41

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1:00 PM–2:00 PM	60 min	Lunch break	Lunch	—
2:00 PM–3:15 PM	75 min	Topic 2 — Data Types and Operators	Concepts, demo and discussion	Slides 43–61
3:15 PM–3:30 PM	15 min	Afternoon break	Break	—
3:30 PM–5:45 PM	135 min	Labs 3–4 and Topic 3 — control structures	Guided hands-on and evidence review	Slides 55–79
5:45 PM–6:00 PM	15 min	Day 1 recap and PM digital attendance	Q&A and reflection	—

Training time: 480 minutes (8 hours); lunch excluded.

Day 2 — Reusable code and Pandas data preparation

Time	Duration	Topic / Activity	Method	Slides
9:00 AM–9:15 AM	15 min	Day 1 recap and AM digital attendance	Q&A and reflection	—
9:15 AM–10:45 AM	90 min	Topic 4 — Scripting with Functions and Lambda	Concepts, demo and discussion	Slides 83–101
10:45 AM–11:00 AM	15 min	Morning break	Break	—
11:00 AM–1:00 PM	120 min	Labs 7–8 — reusable ratios and screening	Guided hands-on and evidence review	Slides 95–100
1:00 PM–2:00 PM	60 min	Lunch break	Lunch	—
2:00 PM–3:15 PM	75 min	Topic 5 — Import and Process Finance Data	Concepts, demo and discussion	Slides 102–125
3:15 PM–3:30 PM	15 min	Afternoon break	Break	—
3:30 PM–5:45 PM	135 min	Labs 9–11 — import, selection and data-quality audit	Guided hands-on and evidence review	Slides 114–122
5:45 PM–6:00 PM	15 min	Day 2 recap and PM digital attendance	Q&A and reflection	—

Training time: 480 minutes (8 hours); lunch excluded.

Day 3 — Aggregation, visualisation and financial analysis

Time	Duration	Topic / Activity	Method	Slides
9:00 AM–9:15 AM	15 min	Day 2 recap and AM digital attendance	Q&A and reflection	—
9:15 AM–10:45 AM	90 min	Topic 6 — Aggregate and Visualize Finance Data	Concepts, demo and discussion	Slides 126–147
10:45 AM–11:00 AM	15 min	Morning break	Break	—

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11:00 AM–1:00 PM	120 min	Labs 12–14 — aggregation, returns and technical analysis	Guided hands-on and evidence review	Slides 138–146
1:00 PM–2:00 PM	60 min	Lunch break	Lunch	—
2:00 PM–3:15 PM	75 min	Topic 7 — Analyze Finance Data	Concepts, demo and discussion	Slides 148–177
3:15 PM–3:30 PM	15 min	Afternoon break	Break	—
3:30 PM–4:00 PM	30 min	Labs 15–17 wrap-up and assessment briefing	Guided hands-on and evidence review	Slides 160–168
4:00 PM–5:00 PM	60 min	Written Assessment (WA) — 8 open-ended questions	Individual open-book assessment	—
5:00 PM–6:00 PM	60 min	Practical Performance (PP) — 5 tasks; PM/assessment digital attendance	Individual open-book assessment	—

Training time: 480 minutes (8 hours); lunch excluded.

Topic-by-Topic Delivery Guide

Topic 01 — Introduction to Python Programming · Slides 24–42

Business requirements · finance use cases · environment · first script

- Business first: Define the decision, users, data, output and acceptance criteria before writing code.
- Finance fit: Python connects market data, calculations, visualisation, automation and AI in one ecosystem.
- Reproducibility: A locked environment makes analytical results easier to review, repeat and audit.
- Accountability: The analyst remains responsible for validating inputs, formulas, assumptions and outputs.

Lab 01 — Set Up a Reproducible Finance Project · Slides 36–38

Facilitation focus: A bank analytics team needs every learner to run the same package versions and verify the environment before analysing SGX data. Expected evidence: The script prints the Python, Pandas, NumPy and Matplotlib versions and ends with ENVIRONMENT READY.

Lab 02 — Translate a Finance Requirement into an SGX Price Tool · Slides 39–41

Facilitation focus: A wealth desk wants a repeatable daily price summary for a small basket of Singapore-listed equities. Expected evidence: The report prints five labelled SGX prices to two decimals, handles an invalid ticker without crashing, and all tests pass.

Topic 02 — Data Types and Operators · Slides 43–61

Numbers · strings · lists · tuples · dictionaries · sets · operators

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- Numbers: Use int for counts and float or Decimal for measured financial values.
- Strings: Clean labels, identifiers, dates and report text before matching or presenting them.
- Collections: Lists, tuples, dictionaries and sets organise positions, identifiers and attributes.
- Operators: Arithmetic, comparison and logical operators encode calculations and decisions.

Lab 03 — Build a Loan Amortisation Calculator · Slides 55–57

Facilitation focus: A relationship manager needs a transparent monthly-instalment calculation including the zero-interest staff-loan case. Expected evidence: For $500,000 at 2.6% over 25 years, payment is about $2,268.35; a 0% loan returns principal divided by months.

Lab 04 — Model a Portfolio with Python Collections · Slides 58–60

Facilitation focus: An investment team needs a clean in-memory representation of holdings, identifiers, sector groups and prices. Expected evidence: Every holding has a market value, portfolio weights sum to 100%, and the overlap set is reported without duplicates.

Topic 03 — Problem Solving with Control Structures · Slides 62–82

If/elif/else · loops · range · enumerate · zip · comprehensions

- Branching: Use if/elif/else when a finance rule selects one outcome from alternatives.
- Iteration: Use loops to apply the same logic to positions, periods or scenarios.
- Loop control: break stops a search; continue skips an invalid record without ending the run.
- Comprehensions: Create a transformed collection concisely when the rule remains readable.

Lab 05 — Encode PE and ROE Valuation Rules · Slides 74–76

Facilitation focus: An equity screen must classify securities consistently at defined PE and ROE thresholds. Expected evidence: All exact-boundary tests pass and each recommendation includes the rule that produced it.

Lab 06 — Loop Through a Portfolio and Build Comprehensions · Slides 77–79

Facilitation focus: A portfolio analyst needs valuations for every valid position while malformed records are logged and excluded. Expected evidence: The invalid row is logged once, valid positions reconcile to the portfolio total, and comprehension results match the loop results.

Topic 04 — Scripting with Functions and Lambda · Slides 83–101

Functions · parameters · return values · scope · lambda · map · filter

- Single purpose: A function should perform one understandable analytical responsibility.
- Inputs and outputs: Parameters make assumptions visible; return values keep computation separate from display.
- Reuse: Well-named functions reduce repeated formulas and inconsistent fixes.
- Lambda: Use small anonymous functions for short transformations, not complex business rules.

Lab 07 — Create Reusable Financial Ratio Functions · Slides 95–97

Facilitation focus: A bank reporting team calculates margin, ROE and ROA in several reports and needs one tested implementation. Expected evidence: The sample produces margin about 55.95%, ROE about 18.09% and ROA about 1.53%; tests pass.

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Lab 08 — Screen Customers with Lambda, Map and Filter · Slides 98–100

Facilitation focus: A retail bank wants to transform customer income into affordability ratios and filter applications for manual review. Expected evidence: Lambda and named-function pipelines select the same applicant IDs, and counts reconcile to the input.

Topic 05 — Import and Process Finance Data · Slides 102–125

Pandas · Series · DataFrame · import · filter · slice · clean

- Series: A labelled one-dimensional array, such as closing prices indexed by date.
- DataFrame: A labelled table of rows and columns suitable for finance records and time series.
- Selection: Use labels and conditions deliberately with loc, iloc and boolean masks.
- Data quality: Missing, duplicate, invalid and inconsistent values must be measured before analysis.

Lab 09 — Import and Clean a Finance DataFrame · Slides 114–116

Facilitation focus: A treasury extract contains duplicated dates, missing prices and inconsistent date/value types. Expected evidence: The curated file has unique sorted dates, numeric prices, no missing Close values, and a reconciliation from raw to clean rows.

Lab 10 — Filter and Slice Finance Data Safely · Slides 117–119

Facilitation focus: A portfolio manager asks for bank-sector positions above a materiality threshold and a date-windowed price series. Expected evidence: The filter returns only qualifying bank positions; loc and iloc row counts match their documented boundary rules.

Lab 11 — Run a Finance Data-Quality Audit · Slides 120–122

Facilitation focus: A credit portfolio extract must pass completeness, uniqueness, range and reconciliation controls before modelling. Expected evidence: The audit identifies every planted defect, produces machine-readable counts and leaves the raw file unchanged.

Topic 06 — Aggregate and Visualize Finance Data · Slides 126–147

concat · merge · groupby · pivot table · returns · charts

- Combine: concat stacks compatible data; merge joins tables using explicit keys.
- Aggregate: groupby splits records, applies calculations and combines results.
- Reshape: Pivot tables summarise a measure across two or more dimensions.
- Visualise: Choose a chart that matches the analytical question and label units clearly.

Lab 12 — Join, Group and Pivot a Loan Portfolio · Slides 138–140

Facilitation focus: A risk team needs exposure and approval metrics by sector, region and risk band from separate application and reference tables. Expected evidence: No unexpected many-to-many join occurs, unmatched keys are reported, and summary exposure totals equal the source.

Lab 13 — Visualize Daily Returns and Distribution · Slides 141–143

Facilitation focus: An investment committee wants to understand daily price changes and the distribution of returns for DBS. Expected evidence: returns.csv has one fewer non-null row than prices; charts render; summary statistics match Pandas calculations.

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Lab 14 — Build a Moving-Average Technical Analysis · Slides 144–146

Facilitation focus: A quantitative analyst needs an OHLC view and 10/50-day moving averages with objectively detected crossover dates. Expected evidence: The output contains OHLC rows, MA10/MA50 values, at least one crossover date, and a labelled chart; the offline file always works.

Topic 07 — Analyze Finance Data · Slides 148–177

apply · pipe · statistics · correlation · rolling analysis · change tracking

- apply: Run a function across rows or columns when vectorised operations are not practical.
- pipe: Express a readable sequence of named DataFrame transformations.
- Statistics: Describe centre, spread, relationships and unusual observations before drawing conclusions.
- Change over time: Returns, rolling windows and drawdown expose behaviour that levels alone can hide.

Lab 15 — Build a Clean-to-Risk Pipeline with pipe · Slides 160–162

Facilitation focus: A credit-risk team wants the DataFrame transformations to read like an auditable sequence rather than one large notebook cell. Expected evidence: Each stage can be tested alone, the raw DataFrame is unchanged, and the final risk-band counts reconcile to total rows.

Lab 16 — Analyse Statistics, Correlation and Rolling Risk · Slides 163–165

Facilitation focus: A portfolio risk analyst must profile returns, compare bank-stock relationships and track changing volatility over time. Expected evidence: Correlation is symmetric with 1 on the diagonal, rolling volatility has the expected warm-up NaNs, and maximum drawdown is non-positive.

Lab 17 — Capstone: Produce an Executive Finance Analytics Pack · Slides 166–168

Facilitation focus: The CFO requests a concise, reproducible evidence pack combining data quality, portfolio exposure, returns, risk and recommendations. Expected evidence: All required files exist, totals reconcile, charts have labels, the HTML report opens locally, and validate_outputs.py prints CAPSTONE PASS.

Resources Required

- Trainer PPTX and learner slide PDF
- Learner Guide DOCX/PDF
- 17 self-contained lab folders with starter, solution, tests and offline data
- Python 3.11+, uv or venv, Pandas, NumPy and Matplotlib
- LMS/TMS access for courseware, attendance and assessment
- Projector, internet access and learner laptops

Assessment

- Written Assessment (WA) — 8 open-ended short-answer questions (K1–K8), 60 minutes, open book.
- Practical Performance (PP) — 5 practical tasks covering A1–A9, 60 minutes, open book.
- Assessment is conducted on Day 3 from 4:00 PM to 6:00 PM.

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- Both instruments are open book and completed individually.
- Candidate papers are submitted on https://lms-tms.tertiaryinfotech.com/
- Answer keys remain trainer-only and are not linked to learners.

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